CSA-2001 Fundamentals in AI and ML

Cosmetics Product Sales Predictor

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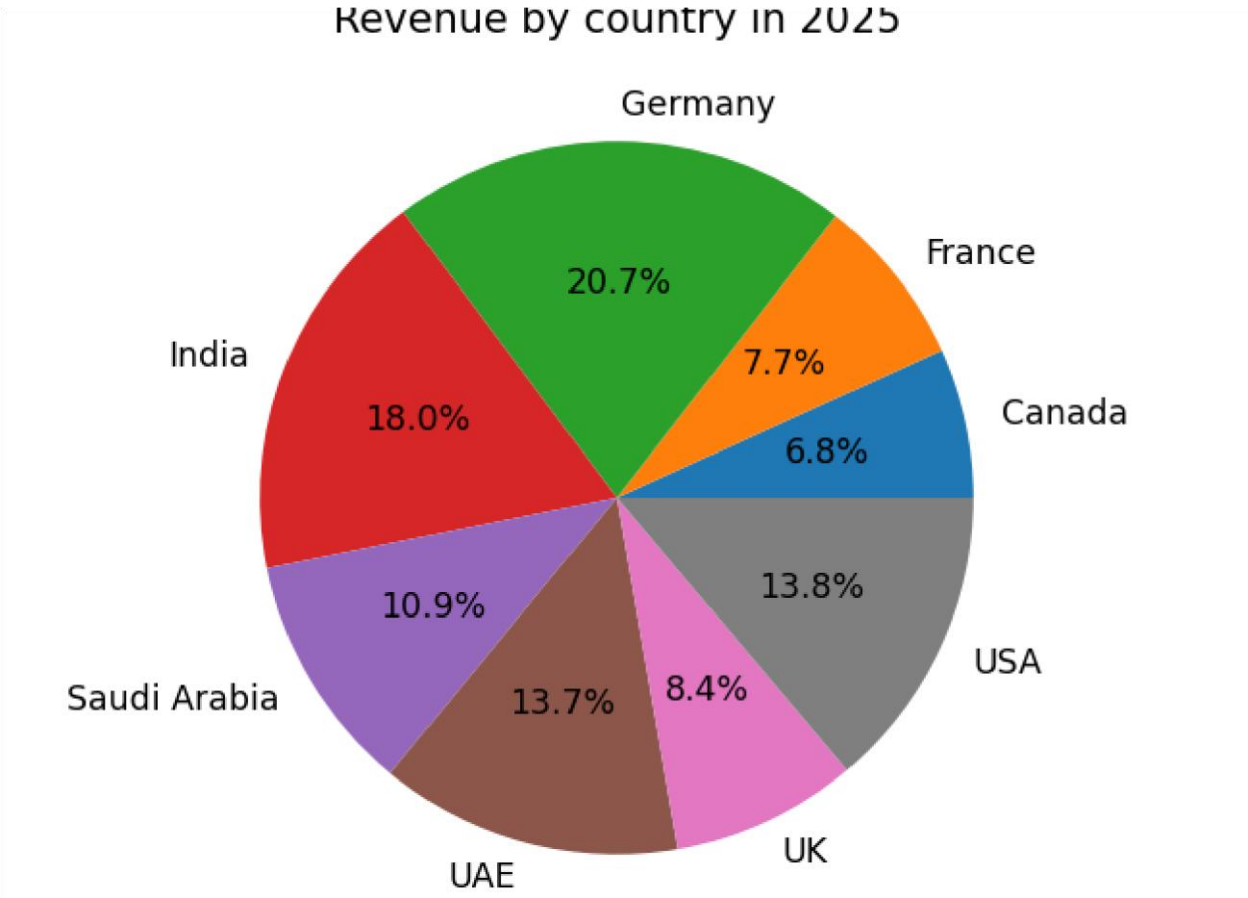

Introduction

This project focuses on analyzing and predicting cosmetic product sales using ML-Linear Regression technique.

The dataset contains information such as brand name, product type, country, sales channel, product's price, units sold, and revenue in United States Dollar.

Cosmetics have a big role in media and lifestyle. Anyone wants to appear attractive and dashing when it comes from a small event to the film industry.

Modules used in the project

- Numpy
- Matplotlib.pyplot
- Pandas
- Sckit-Learn

Trends (According to the Model)

According to the Model, Eyeliner was the player in Cosmetic products.

L'Oreal was the most popular brand throughout the world with its products.

Germany was the largest consumer of cosmetic products during May with 465 units sold.

Model Performance

With Coefficient of Determination(R^2) at 0.8847230123258571, it performed the statistical prediction of the sales of Cosmetic Products.

It had Mean Squared Error of 305266.42401565163 and Root Mean Squared Error of 552.5092071772665, making it an efficient model for predicting the sales.

What I learnt from it?

It was the first time I used the scikit-learn library for my project. It is a vast module which I found challenging at the start but I learned while building the project. Alongside, I used Matplotlib and Numpy for visualising the data.

It was my first application of Machine Learning Technique (Linear Regression) in my academic life.

I made sure that each stone was unturned in the project and I will try my best to learn it in my further path.